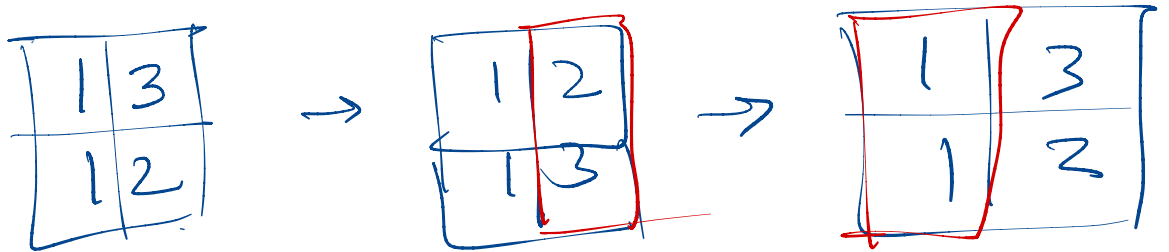


Q: Why is stability required in radix sort?

Example:



Complexity of Radix sort: (on d-digit numbers)

$$T(n, d) = \Theta(d(n+k))$$

Annotations for the complexity formula:

- d : d-digit
- n : input size
- k : base-k numbers

$$T(n, b) = \Theta\left(\frac{b}{r} (n + 2^r)\right)$$

✗ (reject case $r \gg \lg n \Rightarrow$ exponential complexity)
 $r = \lg n$

$$\begin{aligned} \Rightarrow T(n, b) &= \Theta\left(\frac{b}{\lg n} (n + n)\right) \\ &= \Theta\left(\frac{bn}{\lg n}\right) \end{aligned}$$

Bucket Sort

→ Assumes that number in input array $A[1:n]$ are sampled from a uniform & independent distribution

→ All numbers in $A[1:n]$ are s.t.

$$A[i] \in [0, 1)$$

IDEA: Divide the interval $[0, 1)$ into n equal sized subintervals.
(Buckets)

→ distribute n input n into these buckets

→ We do not expect many numbers falling into each bucket

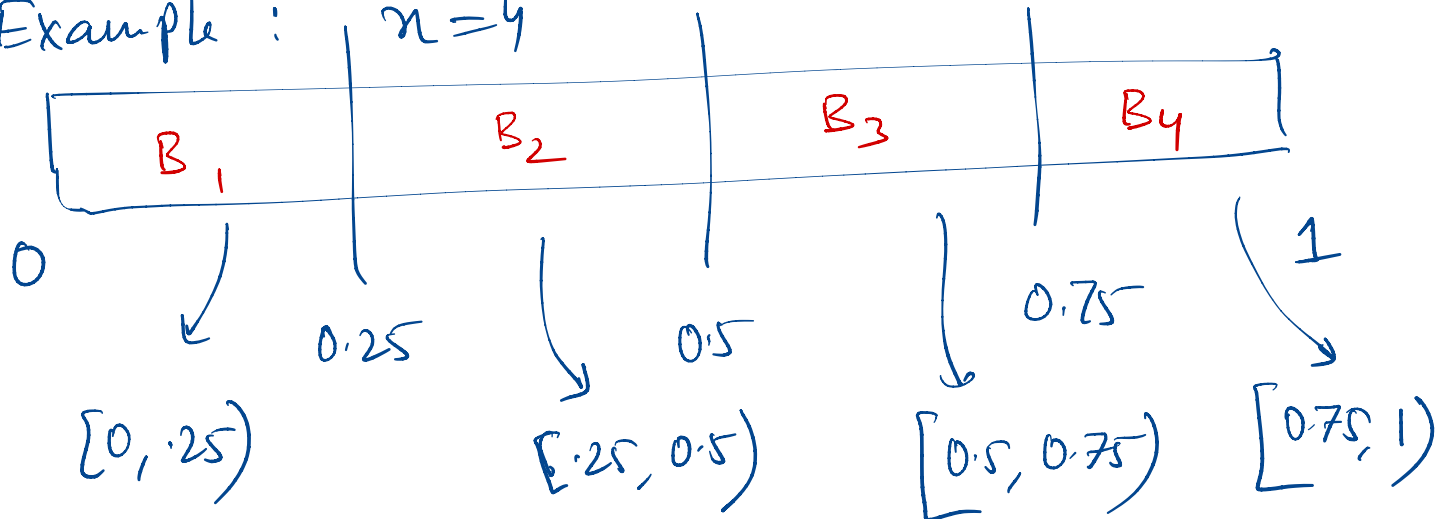
→ To produce the output:

a) sort numbers in each bucket

b) go through each bucket in order listing each element

→ gives the sorted sequence

Example : $n=4$



A : $\begin{bmatrix} 0.99 \\ 0.17 \\ 0.01 \\ 0.81 \end{bmatrix}$ $\rightarrow$ B_4
 $\rightarrow$ B_1
 $\rightarrow$ B_1
 $\rightarrow$ B_4

$\overbrace{0.01, 0.17}^{B_1}, \overbrace{0.81, 0.99}^{B_4}$

$\rightarrow$ Sorted Seq.

$\rightarrow$ need sorting
 $B_1 : [0.17, 0.01]$
 $B_2 :$
 $B_3 :$
 $B_4 : [0.99, 0.81]$

$\Downarrow$

$B_1 : 0.01, 0.17$
 $B_2 :$
 $B_3 :$
 $B_4 : 0.81, 0.99$

$\Leftarrow$

[Input Array : $A[1:n]$ when $A[i] \in [0,1)$
create an array $B[0:n-1]$ of linked lists (buckets)

Bucket - Sort (A, n)

Let $B[0:n-1]$ be a new array

for $i = 0$ to $n-1$

make $B[i]$ an empty linked list

for $i = 1$ to n

Insert $A[i]$ into list $B[\underbrace{\lfloor n \cdot A[i] \rfloor}_{\text{floor}}]$

for $i = 0$ to $n-1$

Sort list $B[i]$ with insertion sort

Concatenate lists $B[0], B[1], \dots, B[n-1]$
together in order

return concatenated list

Time complexity of Bucket sort

$$T(n) = \Theta(n) + \sum_{i=0}^{n-1} O(n_i^2)$$

n_i : # of elements in bucket $B[i]$

All loops are linear time except the
sorty loop (?)

Average-case analysis

$$E(T(n)) = E\left[\Theta(n) + \sum_{i=0}^{n-1} O(n_i^2)\right]$$

$$= \Theta(n) + E\left[\sum_{i=0}^{n-1} O(n_i^2)\right]$$

$$= \Theta(n) + \sum_{i=0}^{n-1} E[O(n_i^2)]$$

... Linearity of expect.

Claim: $E[n_i^2] = 2 - 1/n$

— for each $i = 0, 1, 2, \dots, n$, each bucket i has the same value of $E[n_i^2]$

$\Rightarrow$ each value from A is equally likely to fall into a bucket

— view each random variable n_i as the # of successes in n Bernoulli trials

Success is when an element goes into bucket $B[i]$

Prob. of Success: $P = 1/n$

Prob. of failure: $q = 1 - 1/n$

$$E[n_i] = n \cdot p = n(1/n) = 1$$

$$\text{Var}[n_i] = n \cdot p \cdot q = (1 - 1/n)$$

$$E[n_i^2] = \text{Var}[n_i] + E^2[n_i]$$

$$= (1 - 1/n) + 1^2 = (2 - 1/n)$$

$$\Rightarrow T(n) = \Theta(n) + n \cdot O(2 - 1/n) = \boxed{\Theta(n)}$$